Gioele Rora

Computer Engineering Student — Embedded Systems, Robotics & Full-Stack Software

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LinkedIn: linkedin.com/in/gioelerora | **GitHub:** github.com/GioeleRora | **Portfolio:** gioggio.vercel.app | **Live Project:** route2025.vercel.app

HONOR SOCIETY	HACKATHON WINNER	ACADEMIC MERITS	ENGLISH TRACK
Selected Member & Inductee-Elect, **IEEE-Eta Kappa Nu (Mu Nu)**	**1st Place @ HackTurin** (EGG Edge-AI Smart City Mesh Node)	Computer Science (**30L/30**), Digital Electronics (**30/30**)	B.Sc. Taught 100% in English; **Cambridge B2 Certified**

Education

Politecnico di Torino — Turin, Italy
B.Sc. in Computer Engineering (Taught entirely in English) — *Sept. 2024 – Present*

- **Honors:** Selected Member & Inductee-Elect, **IEEE-Eta Kappa Nu (Mu Nu Chapter)** — IEEE Honor Society.
- **Relevant & Key Coursework:** Computer Science Principles (30L/30), Digital Electronics Design (30/30), Mathematical Analysis I & II, Linear Algebra & Geometry, Physics.
- **Language Certification:** Certified English proficiency level B2 (Cambridge English Assessment).

Liceo Scientifico (Applied Sciences) — Alba, Italy
Diploma di Maturità — Final Grade: 80/100 — *Sept. 2019 – July 2024*

- Active participant in individual and team-based Mathematics, Physics, and Informatics Olympiad competitions.

Fondazione Golinelli — Summer Science Academy — Italy
Interdisciplinary Biotech & Electronics Program — *July 2022*

- Engineered an Arduino-based automated thermocycler, integrating hardware microcontrollers with experimental biological protocols.

Engineering Experience, Work & Student Teams

Team Astra — Politecnico di Torino — Sept. 2024 – Present
Embedded Firmware Engineer — IT Division (Spherical Lunar Rover) — *Turin, Italy*

- Collaborating in the IT division for an autonomous spherical lunar exploration rover powered by internal pendulum motion and robotic arm manipulation.
- Developing, profiling, and testing embedded C/C++ firmware for DC motor velocity regulation, PWM control, and environmental temperature sensor integration.
- Optimizing and refactoring motor management driver libraries for resource-constrained microcontrollers; successfully validated actuation systems for public technical exhibitions.

Team QubiTo — Politecnico di Torino — Oct. 2026 – Present
Software Division Member — Quantum Computing Student Team — *Turin, Italy*

- Collaborating on quantum algorithm design, exploring quantum circuit optimization paradigms and simulation benchmarks.

STEM Private Tutor — 2021 – Present
Independent Tutor in Mathematics, Physics, and Computer Science — *Turin, Italy*

- Instructing high school and early college students, translating complex theoretical formulas into structured, practical, algorithmic problem-solving methodologies.

Technical Projects & Hackathons

EGG — Smart City Mesh Network (1st Place @ HackTurin) | *Zephyr RTOS, C++, Python, Docker, Arduino UNO Q* Oct. 2026

- Spearheaded 3D enclosure engineering and led the team to **1st place**, co-architecting a ~€100 Edge-AI sensor node to replace €15k–€25k proprietary intersection hardware with an urban mesh network.
- Architected an asymmetric dual-core embedded pipeline on Arduino UNO Q: offloaded hard real-time I2C sensor polling and traffic light state-machines to **Zephyr RTOS (C++)**, communicating via RPC (<7 ms latency) with an MPU running Debian/Docker.
- Integrated on-device YOLOX-nano vision models on an 8MP Sony IMX219 camera for GDPR-compliant inference, broadcasting live pedestrian, traffic flow, and crash triage alerts to dynamic signage and a 3D web map.

Drone Delivery Simulator — 3D Aerodynamics & Systems Game | *Unity, C#, 3D Physics Engine, HLSL / Shader Graph, Git* Sept. 2026 – Present

- Engineering a physics-based flight simulator in Unity featuring real-time payload mass shifts, drag coefficient calculations, and dynamic battery/thermal drain systems.
- Architecting procedural mission generation pipelines combined with branching narrative arcs, state and karma-driven faction logic, and unlocking mechanics for vertical level exploration.
- Designing a low-latency analog gamepad input framework and continuous, loading-free level deployment using custom modern glassmorphism UI shaders.

Route 2025 — Full-Stack Event Logistics Platform | *Node.js, JavaScript (ES6+), HTML5/CSS3, Vercel, Git CI/CD* | **Live Demo: route2025.vercel.app** Jan. 2025 – Present

- Engineered a centralized management platform for Scout event logistics, optimizing live attendance tracking, scheduling, and communication for multiple regional groups.
- Leveraged AI-assisted agile development to optimize code modularity and deliver a responsive, mobile-first administrative dashboard.
- Implemented modular REST endpoints and maintained automated CI/CD deployment pipelines via Vercel with continuous integration and branch protection on GitHub.

Leadership, Volunteering & International Experience

Unit Leader (Capo Reparto) — AGESCI Scout Group Sept. 2025 – Present
Youth Leadership & National Event Logistics *Alba, Italy*

- Mentoring and directing a youth group of 30+ individuals; planning yearly educational pathways, managing emergency risk protocols, and executing complex national route logistics.

Erasmus+ Cultural & Academic Exchange Feb. 2023
International STEM & Intercultural Program *Opava, Czech Republic*

- Participated in an intensive international academic program, collaborating with European peers on STEM workshops and intercultural seminars.

Volunteer Worker — Operazione Mato Grosso (OMG) 2015 – Present
High-Altitude Infrastructure & Shelter Maintenance *Rifugio delle Marmotte, Italy*

- Supported infrastructure building, structural renovation, and operational maintenance at Rifugio delle Marmotte in high-altitude alpine environments.

Technical Skills, Languages & Personal Interests

Programming Languages	C, C++, C#, Python, Java, JavaScript (ES6+), HTML5/CSS3, SQL
Systems, Embedded & Hardware	Linux (Debian, CLI), Zephyr RTOS, Embedded C/C++, STM32, Arduino (UNO Q), DC Motor Control & Actuators, Docker, Edge-AI / YOLOX, I2C/SPI, PWM
Developer Tools & Frameworks	Git, GitHub Actions (CI/CD), Unity Engine, 3D Physics Engine, HLSL / Shader Graph, Node.js, Vercel, SQLite
Spoken Languages, Interests & Info	**Languages:** Italian (Native), English (B2 Certified — Cambridge English Assessment / Full Professional Working Proficiency)